

Article

A Conceptual Technical–Economic Framework for Life-Cycle Management of Residential Photovoltaic Modules

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Abstract

This study develops a technical–economic model to support planning for second-life reuse and end-of-life management of residential PV modules in Brazil. The model integrates module degradation, residential electricity-demand dynamics, and consumer utility-based replacement decisions to identify temporal milestones at which modules remain technically functional but become vulnerable to premature replacement. By combining technical performance with consumer behavior, the model captures critical transition points between continued operation, replacement, and reuse pathways. Simulation results indicate that premature PV waste generation is highly sensitive to household consumption growth. Under early-growth demand profiles, users reach the attention and break-even milestones at 11.7 and 13.7 years, respectively, whereas under stable-growth profiles, these thresholds occur only at 27.6 and 29.9 years. This difference substantially increases the period during which modules may be replaced before reaching the estimated technical end-of-functional-life threshold of 29.6 years, demonstrating how demand-side dynamics can accelerate waste generation independently of module degradation. The findings indicate that integrating technical and economic criteria into planning decisions may help identify opportunities to extend module service life and reduce avoidable PV waste. The proposed model contributes to the waste-management literature by providing a scalable decision-support basis for reuse assessment, reverse logistics planning, recycling prioritization, and evidence-based policy development. More broadly, it may contribute to strategies that connect low-carbon energy transitions with circular economy principles and the United Nations Sustainable Development Goals (SDGs).

Keywords: second-life; waste management planning; premature disposal



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1. Introduction

Electronic waste has become a growing environmental concern, not only because of the significant increase in its volume in recent years but also due to the risks it poses to human health and the environment when improperly disposed of [1,2]. This category of

waste includes electronic equipment discarded due to obsolescence, functional failure, or disuse. In the context of the energy transition, waste generated from photovoltaic (PV) systems has gained prominence, as the rapid global expansion of this technology imposes significant logistical, regulatory, and environmental challenges, particularly regarding the management of PV modules at the end of their service life [3–5].

It is estimated that, by 2050, more than 78 million tons of PV panel waste will be generated globally, as shown in Figure 1 [6]. This growing volume can be directly associated with the widespread adoption of solar PV energy as a clean and low-cost alternative, whose competitiveness has been driven by continuous technological advances [7–9]. Projections suggest that, by 2050, the installed capacity of these systems may increase by more than 1800% compared with current levels [6,10]. Meanwhile, the ongoing modernization of PV modules has accelerated the early replacement of equipment that remains operational, whether because of gradual efficiency losses or technological obsolescence relative to newer, higher-performing models [11]. This phenomenon directly contributes to the growing volume of waste, even when replaced modules still retain residual technical capacity.

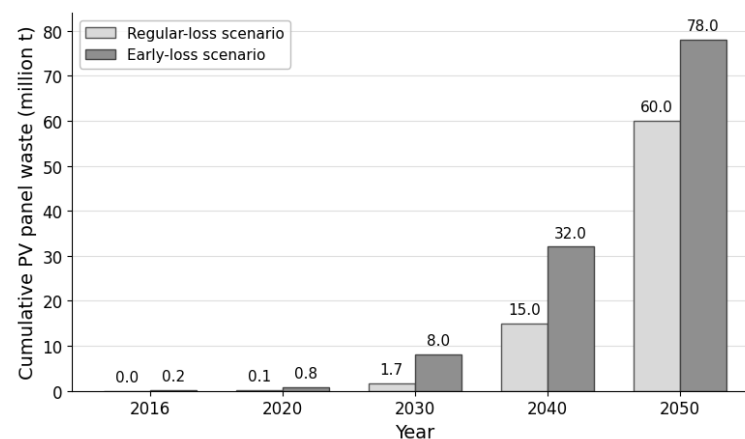


Figure 1. Projected cumulative global PV panel waste, 2016–2050. Adapted from [6].

From an end-of-life (EoL) management perspective, PV modules can follow three primary pathways: disposal, recycling, and reuse [12,13]. Among these, disposal, typically via landfilling or incineration, represents the least desirable option, as it leads to the permanent loss of valuable materials and poses environmental risks if not properly managed [14–16]. Inadequate EoL governance may result in the release of hazardous substances and long-term soil and water contamination, reinforcing the need for regulatory frameworks and incentive mechanisms to prevent uncontrolled disposal [14,16].

On the other hand, recycling is often considered a definitive solution, as it enables the recovery of materials such as glass, silicon, and valuable metals, thereby reducing the need for virgin resource extraction [12,15,16]. However, despite these advantages, current recycling processes face limitations. The complex structure of PV modules makes dismantling and material separation technologically challenging and energy-intensive [17–19]. In addition, recycling operations may generate secondary waste streams, including contaminated residues that require controlled disposal, often in landfills, due to their potential environmental risks [14,15]. Economic and regulatory barriers continue to limit large-scale viability, while environmental concerns persist regarding energy consumption, emissions and the release of potentially toxic substances during processing and transportation [17,18,20].

In contrast, second-life applications (reuse strategies) have gained attention as an intermediate yet effective means of extending the functional lifespan of PV modules [21]. Although reuse does not eliminate the eventual need for recycling, it delays EoL processing and preserves the embedded value of existing systems, thereby reducing immediate waste

generation and the demand for new module production [12,15,16]. This delay can also help minimize the cumulative generation of secondary residues associated with recycling processes [22]. This delay can also help minimize the cumulative generation of secondary residues associated with recycling processes [11,22,23]. Despite being a transitional solution, reuse plays a critical role in circular-economy strategies by bridging the gap between initial use and final material recovery while alleviating pressure on resource extraction and waste-management systems [12,13,24].

Despite international progress in discussions on EoL management and circular economy practices in the PV sector, significant technical and scientific gaps persist, especially in emerging contexts where institutional constraints, high capital costs, and regulatory delays exacerbate these challenges [11,23,25]. From a technical-scientific standpoint, the absence of standardized guidelines for testing, qualifying, and reintegrating used modules into new systems compromises both scalability and confidence in commercial reuse [11,26]. Furthermore, empirical studies on repair and disposal options remain limited, particularly those integrating technical, economic, and behavioral variables within a unified analytical framework. From a microeconomic perspective of consumer behavior, it is believed that perceptions of residual utility and the risks associated with module reuse can influence premature replacement decisions, even when technical operating potential remains.

Some studies reinforce the relevance of different dimensions of PV module life-cycle management (LCM). Aboagye et al. (2022) [27] analyze the degradation behavior of PV module technologies under different climatic conditions, highlighting the importance of technical performance assessment for estimating service life. Van der Heide et al. (2022) [11] discuss requirements for the successful reuse of decommissioned PV modules, including testing, reliability, and reintegration challenges. Complementarily, Marinna et al. (2025) [23] propose a decision-making framework for PV module reuse within a circular solar economy perspective. Together, these contributions show that PV LCM involves technical degradation, reuse qualification, reconditioning pathways, and circular-economy decision processes.

Economic decision-making models provide a complementary perspective. Moon and Baran (2018) [28] develop a real-option model for determining the optimal timing of residential PV investment under system-cost uncertainty and the option to defer adoption. Wang et al. (2022) [29] formulate sequential real-option decisions for PV projects under renewable portfolio standards and tradable green certificate markets, addressing optimal investment timing, installed capacity, and phased investment. In the Brazilian residential context, Leite et al. (2024) [30] apply discounted cash-flow indicators to compare PV investment performance across net-metering rules, household consumption levels, and discount rates. These studies demonstrate that PV decisions are sensitive to uncertainty, project scale, electricity-consumption profiles, and policy incentives. However, they primarily examine initial adoption, investment timing, or project expansion and do not jointly represent post-installation module degradation, changing household demand, and perceived residual utility in decisions concerning continued use, replacement, or second-life allocation.

Building on these technical and economic contributions, the present study focuses on the residential owner perspective by integrating module degradation, energy-consumption growth, and perceived utility as decision-support elements for assessing the second useful life of PV modules. Thus, the main contribution of this work does not lie in proposing a new degradation model, investment-valuation method, or recycling strategy, but rather in coupling technical degradation, household demand evolution, and owner-oriented utility criteria within a unified framework designed to support PV module LCM.

Given this scenario, this exploratory and applied study proposes a conceptual decision-support framework for simulating dynamic scenarios, grounded in technical and microeconomic literature on consumer behavior, to examine how PV module degradation, energy

consumption patterns, and perceived utility may influence reuse decisions in Brazilian residential applications. The proposed framework was based on the principles of utility theory and diminishing consumer satisfaction, enabling the exploration of potential interactions between technical performance and economic decision-making over time. Accordingly, different usage and replacement scenarios are constructed based on real-world parameters of efficiency, cost, and consumption, in order to assess the potential for extending module usage and mitigating premature disposal.

2. Theoretical Framework

The debate surrounding the life cycle of PV modules has intensified as the global volume of installed equipment continues to rise and, consequently, the potential waste generated at the end of their service life increases. Within the context of the circular economy, the reuse of PV modules stands out in comparison to recycling, as it preserves the functionality of the equipment, reduces the energy consumption associated with EoL industrial processes, and aligns with the core principles of circularity [12]. Moreover, it prevents the loss of valuable materials such as silicon, silver, and indium, and can contribute to lowering the Levelized Cost of Energy (LCOE) by shortening the payback period of the initial investment [10,19,31]. However, the current absence of technical standards for qualifying and reintroducing used modules, along with the scarcity of circular business models and the absence of specific regulatory guidelines, still constrains the large-scale adoption of reuse practices [3,11,23,25,32].

The reuse of technologically obsolete modules emerges as a sustainable and economically viable alternative, particularly in low-demand contexts such as isolated communities, small rural producers, social projects, and academic institutions [33–35]. By preserving equipment functionality and mitigating the costs associated with purchasing new systems, reuse promotes the democratization of access to clean energy and reduces the environmental impacts linked to the premature disposal of still-operational modules [24,36]. This approach also reinforces the alignment among sustainability, energy inclusion, and technological innovation, key elements of public policy in developing countries.

From a technical perspective, two main factors explain the growing volume of discarded PV modules: technological obsolescence and natural degradation over time [6,37,38]. Obsolescence occurs as newer, more powerful, and efficient models make previous versions less competitive, even if they remain operational [11,27,39]. Degradation, in turn, refers to the progressive loss of efficiency caused by climatic factors, operating time, and material characteristics, directly affecting the service life of the modules, typically estimated between 25 and 30 years, with degradation rates ranging from 0.58% to 0.83% per year [6,27,37–41]. Generally, modules are considered to have reached the end of their technical life once their power output falls to approximately 80% of their original rated capacity, a criterion widely used in technical references from the U.S. National Renewable Energy Laboratory (NREL), a research laboratory specialized in renewable energy technologies [42].

In this context, and given the accelerated growth of PV installations in recent decades, concerns have intensified regarding the volume of materials, such as glass, ethylene-vinyl acetate (EVA), and aluminum, that will be generated at the end of this cycle, highlighting the urgency of strategies aimed at extending the operational life of PV equipment [3,6,23,32,43]. Beyond these technical aspects, institutional, economic, and market variables also contribute to premature replacement, including regulatory changes, modernization incentives, and declining costs of new modules. These factors reinforce the need to better understand the replacement decision-making process [11,44].

Emerging economies may particularly benefit from PV module reuse mechanisms, given the persistently high costs of installing new solar systems [45]. This remains a signif-

icant barrier to the widespread diffusion of the technology, especially in middle-income or developing countries such as Brazil, which has a long-standing tradition of renewable energy adoption. In the Brazilian context, the Brazilian Electric Energy Agency (Agência Nacional de Energia Elétrica, ANEEL), the federal regulatory agency responsible for regulating the electricity sector, established Normative Resolution No. 482/2012, which created the regulatory framework for micro- and mini-distributed generation and contributed to the subsequent expansion of PV systems in the country [46]. The cost of electricity accounts for a considerable portion of household finances, often requiring public subsidy programs to ensure equitable access to electricity, particularly among socioeconomically vulnerable families [47].

The literature indicates that the feasibility of reuse depends not only on technical and regulatory parameters but also on consumers' economic and behavioral perceptions. From a microeconomic perspective, replacement and reuse decisions are driven by perceived utility, relative costs, and perceived risks associated with refurbished equipment. Within this framework, the decision to reuse can be conceptualized as an intertemporal utility maximization problem, in which the consumer compares the energetic and financial benefits of the current module against the opportunity cost of replacement. Marginal utility decreases as technical performance declines, influenced by factors such as income, risk aversion, and investment time horizon [48,49].

Recent studies reinforce this framework. Koide et al. (2025) [50] identify, through conjoint analysis and market simulations, that price, reliability, risk perception, and environmental engagement shape consumer preferences for circular products. Complementarily, Aydin and Mansour (2023) [51] demonstrate that the willingness to purchase remanufactured products was strongly associated with perceived utility and confidence in technical performance. These findings align with evidence suggesting that reuse decisions involve economic, psychological, and symbolic components. These findings align with evidence suggesting that reuse decisions involve economic, psychological, and symbolic components, indicating that consumer rationality can be bounded and shaped by subjective perceptions of performance and return [52,53].

The literature on PV module LCM addresses several complementary dimensions. Technical performance and degradation have been analyzed to estimate service life under different operating and climatic conditions [27,39,40]. Other studies focus on recycling processes, material recovery, and environmental impacts associated with PV waste treatment [12,14,18]. Reconditioning, reuse qualification, reverse logistics, and circular-economy pathways have also been discussed as mechanisms for extending module lifetime and reducing premature waste generation [11,13,23]. Building on these contributions, the present study examines a complementary dimension: the interaction between residential production–consumption dynamics and end-user-oriented decision-making. In this context, linking module degradation, household electricity-demand growth, and decision-support criteria within a single analytical structure can help identify cases in which modules still retain technical capacity, but changes in household consumption modify the perceived usefulness of the system and motivate maintenance, expansion, replacement, or reuse decisions.

Based on this theoretical and empirical foundation, the next section presents the methodological framework adopted to simulate dynamic scenarios of PV module reuse in Brazilian residential systems. The proposed conceptual framework combines technical parameters of degradation and efficiency with economic and behavioral variables, enabling an exploratory analysis of how replacement and reuse decisions may evolve over time under different market conditions and consumer profiles. The main innovation of the study does not lie in developing a new degradation model or proposing a new PV waste-management

strategy, but rather in integrating module degradation, household-consumption evolution, and decision-support criteria into a single technical–economic structure oriented toward residential consumers. This integration supports the identification of temporal milestones throughout the PV module life cycle, offering a methodological basis that can assist decisions regarding continued operation, system expansion, replacement, reuse, or end-of-life planning.

3. Materials and Methods

The methodology adopted in this study was designed to reflect the Brazilian context of PV module reuse, integrating microeconomic and behavioral foundations within the framework of the circular economy. The proposed conceptual framework was structured around three complementary dimensions: technical, economic, and behavioral.

The first dimension, from a technical perspective, addresses the projection of energy generation and consumption from residential PV systems over time. These projections are based on efficiency parameters and average degradation rates reported in the literature and by institutional sources [6,38].

The second dimension, of an economic nature, was grounded in microeconomic consumer theory, particularly in the principles of utility maximization under budget constraints and diminishing marginal utility [48,49]. Replacement can be modelled as an intertemporal choice involving a durable good, in which the PV system provides an energy service whose output flow is affected by technical degradation. The consumer compares the marginal utility gain from replacing or upgrading the module with the replacement cost and income constraint, seeking the break-even at which the marginal benefit of reuse equals the marginal cost of substitution.

The third dimension integrates the technical and economic perspectives to identify the intersection between system performance and diminishing consumer satisfaction. By combining estimates of energy production, consumption patterns, and consumer satisfaction levels, the model enables the construction of simulated scenarios that reveal the potential socioeconomic impacts of PV module reuse.

Accordingly, this research was exploratory and applied in nature, adopting a quantitative approach aimed at analyzing the interactions among technical performance, economic value, and consumer behavior, while dynamically representing reuse decisions within the Brazilian residential context.

3.1. Projected Energy Consumption and Production

Energy consumption was characterized based on data from the Energy Research Office (EPE), which presents per capita information organized by social class [54]. Although useful for preliminary estimates, average household consumption varies significantly across regions and income groups. Moreover, factors such as increasing electrification, home automation, digitalization and Internet of Things (IoT), and the adoption of electric vehicles have been reshaping energy consumption patterns [55]. Thus, while income remains one of the main determinants of consumption, it does not, by itself, fully explain energy demand. Other relevant factors include advertising, which encourages the replacement of outdated appliances with more efficient models and promotes new consumption habits [56], as well as behavioral aspects, such as individual preferences, lifestyles, and the early adoption of technologies by specific groups [57].

These variables make the estimation of consumption behavior complex and subject to uncertainty. To better capture this heterogeneity, segmentation by social class was adopted, since energy behavior can be directly associated with socioeconomic conditions,

particularly household income, which influences both the quantity and the pattern of electricity use [58,59].

Consumption was related to social classes and residential appliance use, following an exponential growth model over time, suitable for cumulative phenomena [60]. Equation (1) expresses this evolution:

$$C(t) = C_0 * (1 + i)^t \quad (1)$$

where:

- $C(t)$: Consumption in year t , in kilowatt-hours;
- C_0 : Initial consumption, in kilowatt-hours;
- i : Annual growth rate, in percentage (%);
- t : Time, in years.

The annual system energy output of a PV system over time (P_{kWh}) can be estimated by Equation (2), which considers three parameters: (i) nominal power; (ii) system configuration; and (iii) production decay

$$P_{kWh} = N * h * 365 * FF * \eta * A * e^{-\lambda * t} \quad (2)$$

where:

- N : Number of panels in the system;
- h : Average daily solar irradiance hours of the system;
- 365: Conversion factor from days to years;
- FF : Fill factor, i.e., percentage (%) of the module area occupied by cells;
- η : Energy conversion efficiency, in percentage (%);
- A : Panel area, in square meters;
- λ : Component failure rate per year, related to module degradation;
- t : Time, in years.

Nominal power depends on the fill factor (FF), the efficiency of solar-to-electric energy conversion (η), and the total module area (A) [61]. The η was determined through standard tests conducted under an irradiance of 1000 W/m² and a temperature of 25 °C [37,61]. System configuration considers the number of modules (N) and the average number of daily irradiance hours (h). Based on these parameters, widely used equations are applied for PV system sizing and computational modelling of energy generation [62,63].

To represent module degradation in production, a reliability function was employed to estimate the residual performance of a non-repairable module, which can be discarded and replaced once failure occurs [64–67]. Equation (3) expresses the reliability of a replaceable component as a percentage. To estimate residual module performance, it was necessary to know the failure rate (λ) or the Mean Time To Failure ($MTTF$), as well as the time (t). With this information, energy production over time can be estimated considering module degradation, as described in Equation (2).

$$R(t) = e^{-\lambda * t} = e^{-\frac{t}{MTTF}} \quad (3)$$

where:

- $R(t)$: Estimated module performance at time t , in percentage (%);
- λ : Failure rate on a time scale;
- $MTTF$: Mean time to failure, in years.

Constant Degradation Assumption and Service-Life Equivalence

Although there is general consensus that a PV module reaches the end of its useful life when its output declines to approximately 80% of its initial capacity, typically after around 25 years of operation [37,42], annual degradation rates may range from 0.3% to 5%, depending on cell type, climatic conditions, irradiance, temperature, humidity, and operational patterns [27,38,68]. Therefore, the real degradation trajectory of a PV module is not necessarily constant over time and may include non-linear effects associated with early-life degradation, material ageing, local environmental stressors, and operating conditions.

Nevertheless, complete degradation curves representative of the reference case adopted in this study were not available in the consulted literature, and long-term experimental ageing tests were outside the scope of the research. For this reason, the degradation parameter was estimated using an equivalence based on reliability and service-life modelling concepts. In this approach, the 80% residual-generation threshold was adopted as the functional end-of-life reference, following widely used NREL guidelines, and the degradation process was represented by a constant failure rate (λ). This assumption allowed the model to translate available estimates of useful life and end-of-life limits into a tractable annual performance-decay parameter, while preserving the possibility of comparing scenarios with different module classes and consumption trajectories.

Accordingly, the classification proposed by NREL was adopted. This classification groups modules according to material, efficiency, and degradation pattern, enabling the identification of module class and technical references for service life estimation. Since failure rate data (λ) are typically reported as ranges, the geometric mean was used as the central estimator, as it was suitable for representing the asymmetric distributions characteristic of technological durability [69]. Equation (4) describes the calculation of the geometric mean (t_g):

$$t_g = \sqrt{\alpha * \beta} \quad (4)$$

where:

- α : Minimum value of the range, in years;
- β : Maximum value of the range, in years.

Finally, to estimate the number of modules required for the system, the estimated consumption and a projected increase were considered, driven by income growth, greater appliance access, and evolving energy-use patterns [63]. The number of modules (N) was obtained using Equation (5):

$$N = \frac{C_0 * (1 + I)}{h * 365 * FF * \eta * A * e^{-\lambda * t}} \quad (5)$$

where:

- C_0 : Initial energy consumption, in kilowatt-hours;
- I : Projected increase in consumption, in percentage (%);
- h : Effective daily solar irradiance hours considered for generation;
- 365: Conversion factor from days to years;
- FF : Panel fill factor, in percentage (%);
- η : Energy conversion efficiency, in percentage (%);
- A : Panel area, in square meters;
- λ : Failure rate of the panel class, on a time scale;
- t : Time, in years.

3.2. Measuring Consumer Satisfaction over Different Periods of Use

Consumer satisfaction with PV systems can be analyzed through an analogy with utility theory, which describes how the satisfaction derived from consuming a good tends to decline as consumption increases (Equation (6)) [48,49,70]. In mainstream microeconomics, utility represents an ordinal measure of preference over bundles of goods, maximized under technical and budget constraints. In this context, the variation in the marginal utility of a durable good, such as a PV system, can be understood as an intertemporal trajectory of satisfaction, in which the additional benefit of consuming the generated energy decreases as the equipment's performance degrades over time (Equation (8)).

$$U(Q) = a * \ln(Q + 1) \quad (6)$$

where:

- $U(Q)$: Total utility perceived by the consumer;
- Q : Level of adoption or the amount of energy generated by PV systems;
- $a > 0$: Parameter expressing the initial intensity of satisfaction.

The logarithmic specification in Equation (6) was adopted because it is a strictly increasing and concave function, ensuring positive and diminishing marginal utility in accordance with the Law of Diminishing Marginal Utility. Unlike the classical quadratic specification, it does not exhibit a decreasing region of total utility nor imply a satiation point. This characteristic is consistent with the nature of PV energy generation: even though the incremental benefit of additional production decreases over time, additional kilowatt-hours never generate negative utility. Therefore, a quadratic function, which allows total utility to fall and marginal utility to become negative, is not appropriate for representing satisfaction from accumulated PV generation [71].

The marginal utility, obtained as the derivative of $U(Q)$ with respect to Q , was expressed by Equation (7):

$$\frac{\partial U(Q)}{\partial Q} = \frac{a}{Q + 1} \quad (7)$$

The second derivative of $U(Q)$ confirming the decreasing behavior of marginal utility was expressed by Equation (8):

$$\frac{\partial^2 U(Q)}{\partial Q^2} = -\frac{a}{(Q + 1)^2} \quad (8)$$

Figure 2 illustrates the classical quadratic total utility function $U(Q)$ and its corresponding marginal utility $MU(Q)$. In this specification, total utility increases at a decreasing rate until reaching a maximum, after which additional consumption reduces total utility and produces negative marginal utility. The point Q' represents the satiation point: at $Q = Q'$, total utility reaches its maximum and marginal utility equals zero. For $Q < Q'$, marginal utility is positive, whereas for $Q > Q'$, marginal utility becomes negative and total utility declines. This formulation, widely used in introductory microeconomics, is pedagogically valuable because it explicitly displays the satiation point and allows for the possibility of consumer dissatisfaction beyond it. Its inclusion here serves as a conceptual benchmark: the quadratic form represents utility structures in which excess consumption generates negative utility. In contrast, the logarithmic function adopted in this study remains strictly increasing and never yields negative marginal utility, reflecting goods for which additional increments reduce satisfaction only marginally, but never adversely. The quadratic specification shown in the figure is therefore not used in the analytical calculations. The comparison between these two forms clarifies the distinction between utility frameworks that admit negative utility and those that model only diminishing incremental satisfaction.

Together, the curves in Figure 2 visually reinforce the classical microeconomic principles of increasing but concave total utility and diminishing marginal utility that ground the analytical approach used to evaluate PV system performance and consumer satisfaction.

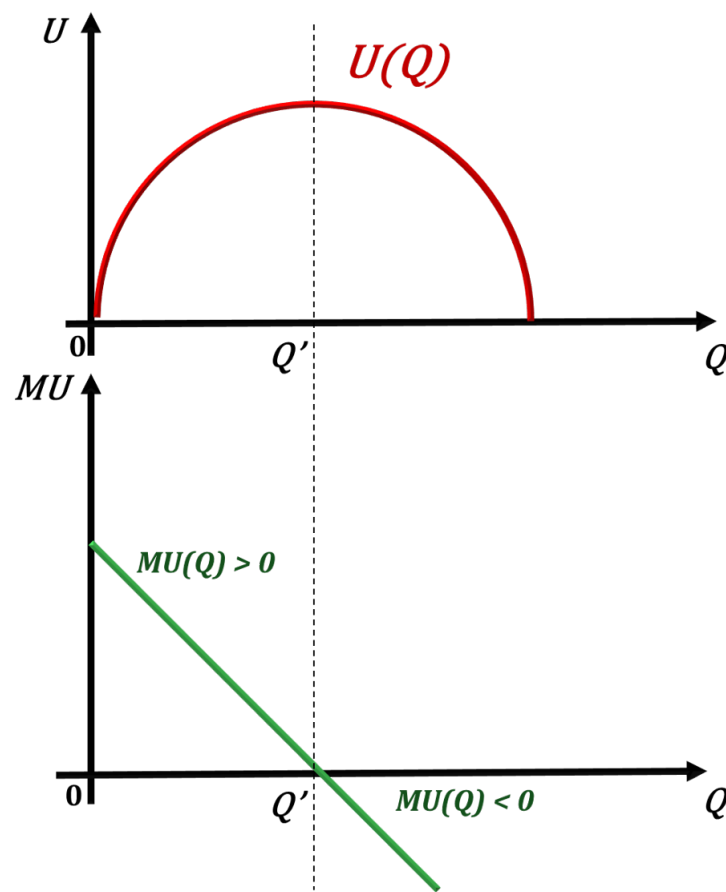


Figure 2. Quadratic total utility and marginal utility functions used as a conceptual benchmark. The total utility function $U(Q)$ increases at a decreasing rate and reaches its maximum at the satiation point Q' , where marginal utility $MU(Q)$ equals zero. Beyond Q' , total utility decreases and marginal utility becomes negative, indicating that additional consumption reduces total utility under the quadratic specification. The red curve represents total utility $U(Q)$. The green curve represents marginal utility $MU(Q)$, which declines as consumption increases and becomes negative after the satiation point, indicating that additional consumption reduces total utility. The vertical dashed line marks the point at which total utility is maximized and marginal utility equals zero, separating the region of positive additional satisfaction from the region of negative marginal utility. Adapted from [71].

Applying this logic to the context of distributed generation, it was assumed that the perceived utility of the system changes over time, influenced both by module degradation and by the relationship between energy production and consumption. The decision to maintain, replace, or reuse the system can therefore be analogous to the classical intertemporal choice problem, in which the consumer compares the expected marginal utility of continued use with the cost associated with replacement, seeking the point where marginal utility equals marginal cost (microeconomic equilibrium condition) [49].

In this study, accumulated behavioral utility (u_1) represents the overall utility derived from the PV system over time, capturing the cumulative benefits perceived by the consumer as electricity generation offsets household demand. More specifically, it reflects the continuous accumulation of the difference between production and consumption ($P - C$), reaching its maximum value and stabilizing when $P = C$, which indicates the break-even point.

Marginal behavioral utility (u_2), in turn, expresses the additional satisfaction obtained from the system at each point in time, that is, the extent to which the system's production capacity continues to meet consumption. This function declines as technical degradation reduces generation capacity and consumption patterns evolve, approaching zero when production becomes lower than consumption ($P < C$), thereby signaling consumer dissatisfaction. While u_1 captures the broader trajectory of perceived value, u_2 highlights the incremental gains or losses associated with the system's continued operation. Together, these functions make it possible to interpret two distinct dynamics: the accumulation of perceived utility up to the break-even point and the subsequent reduction in satisfaction caused by the progressive loss of system efficiency over time.

Based on microeconomic theory, the measurement of these functions reflects the rational behavior of a representative agent seeking to maximize intertemporal utility under technical and economic constraints [48]. To quantify this dynamic, Equations (9) and (10) were used to estimate, respectively, the accumulated utility of the system (u_1) and the perceived utility (u_2):

$$u_1 = \sum_{i=0}^n P_i - C_i \quad (9)$$

$$u_2 = P_i - C_i \quad (10)$$

where:

- u_1 : Total (accumulated) utility of the system in year i , in kilowatt-hours;
- u_2 : Marginal utility of the system in year i , in kilowatt-hours;
- P_i : Energy production of the system in year i , in kilowatt-hours;
- C_i : Energy consumption in year i , in kilowatt-hours.

Thus, the utility function represents the difference between the energy produced and consumed, with marginal utility derived from the concave shape of the function, following the principles of classical microeconomic theory [48,49]. This formulation makes it possible to approximate consumer behavior in the context of distributed generation, linking technical performance with perceived economic utility throughout the system's life cycle.

3.3. Assumptions for Defining Scenario Milestones

To evaluate the different possibilities for the reuse, maintenance, or replacement of PV modules over time, a scenario-based dynamic approach was adopted, integrating both technical and economic perspectives. Defining these behaviors requires identifying milestones that represent significant changes in system performance and consumer satisfaction, thereby guiding both modeling and comparative analysis.

Based on this premise, three technical and economic reference points were established:

- I. The point at which the system begins to show a loss of technical performance and a reduction in user satisfaction;
- II. The moment when energy production equals consumption, marking the end of surplus generation;
- III. The technical limit of the system's functional lifetime.

These reference milestones guide the construction of behaviors and support consumer decision-making, enabling the analysis of reuse behavior under different combinations of technical performance, economic cost, and perceived utility.

4. Results

To estimate energy production and consumption as a function of income, a representative list of household appliances was compiled for each social class, excluding complementary consumption elements such as electric mobility. For each class, the nominal power of household appliances, their average daily usage time, and the average quantity per household were considered to estimate the annual consumption per appliance, and consequently, the total consumption per income bracket. Table 1 presents the social class, annual income range, estimated annual consumption, energy-consumption profile, and the corresponding justifications for each consumption pattern. In addition to estimating consumption, the study sought to identify the energy-behavior pattern for each social class, acknowledging that income, although a determining factor, is not sufficient by itself to explain differences in consumption, which are also influenced by technological and behavioral factors reported in the literature [54,58].

Table 1. Estimated Brazilian residential electricity consumption by social class for 2025. The table summarizes the annual household income range, estimated annual electricity consumption, household energy-consumption profile, and main determinants of consumption for each income group. Values were estimated by the authors from representative household-appliance ownership, power ratings, and usage patterns, and were cross-checked against Brazilian sectoral statistics and income-class consumption references [44,54,72–74].

Social Class	Annual Household Income Range [BRL]	Estimated Annual Consumption [kWh]	Household Energy Consumption Profile	Determinants of Household Energy Consumption
A	Above 240,000	18,428.40	High consumption, stable	Durable goods saturation; inelastic consumption; expansion via automation
B1	120,000 to 240,000	10,173.60	High consumption, expanding	Adoption of higher-power appliances; simultaneous demand increase
B2	60,000 to 119,999	6921.00	Medium consumption, moderate growth	Consolidation of comfort standards; gradual energy use increase
C1	30,000 to 59,999	4696.20	Low-medium, rapid growth	New or upgraded appliances; longer usage time
C2	18,000 to 29,999	2608.20	Low, but increasing	Gradual adoption of microwave devices and washing machines
D/E	Up to 18,000	1381.50	Very low, recent addition of appliances	Recent grid access or improved consumption; rapid growth

Regarding the estimated annual consumption values presented in Table 1, a comparison was made with the Fact Sheet—Residential Energy Consumption by Income Class for validation purposes. The results proved consistent with Brazilian benchmarks [54], even though the estimates do not account for regional or behavioral variations. To estimate consumption behavior, as described by Equation (1), Class A (Table 1) was adopted as the reference group. This choice does not imply that lower-income consumers are less relevant for second-life PV markets; on the contrary, these groups may represent an important future demand segment because refurbished modules can reduce acquisition costs and support broader access to distributed generation. The selection of Class A was motivated by the historical evolution of the Brazilian distributed-generation market after ANEEL Normative Resolution No. 482/2012, when PV systems still had high upfront costs and spe-

cific financing instruments were limited, concentrating early adoption among households with higher purchasing power [46,72]. Therefore, Class A was used as a reference scenario representative of the initial expansion phase of residential PV generation in Brazil and because it presents the highest estimated annual electricity consumption among the income groups shown in Table 1, particularly in the column “Estimated Annual Consumption [kWh]”, rather than as a fixed socioeconomic target group. Growth rates were defined parametrically, within plausible ranges observed in the literature, considering the influence of factors such as income, public policies, technological innovation, access to new appliances, and tariff variations [75,76]. In this context, it is important to emphasize that the framework is not restricted to a specific socioeconomic group and can be adapted to any Brazilian income bracket by adjusting the reference annual-consumption value and the associated electricity-demand growth parameters.

The proposed framework was developed as a parameterizable and adaptable simulation structure, using technical parameters and reference values obtained from official databases, technical standards, and specialized literature; consequently, the same structure can be applied to other income groups, regions, and PV module technologies by adjusting the input parameters for consumption, demand growth, system size, irradiation conditions, module specifications, and decision thresholds. These sources include the useful-life criteria for PV modules established by NREL, solar irradiation data from the Brazilian Solar Energy Atlas produced by the Brazilian Institute for Space Research (INPE), evidence from studies on Brazilian climate and solar-resource conditions, and residential electricity-consumption estimates based on official Brazilian statistics and sectoral reports [42,54,77–79]. Therefore, the objective of the simulation framework is not to reproduce a single observed PV installation or to restrict the analysis to one socioeconomic group, region, or commercial module, but to explore how technical degradation and consumer behavior interact under different electricity-consumption growth scenarios.

To estimate energy production, as described by Equation (2), we considered technical parameters representative of photovoltaic modules currently available in the Brazilian market. Because consolidated technical yearbooks on solar panel sales in Brazil are not available, representative module specifications were compiled from sectoral sources and specialized market reports. Since PV generation is inherently variable and depends on seasonal, geographic, and climatic factors such as solar irradiance and cloud cover, energy production does not remain constant over time and may not match household consumption profiles. Based on the lifetime ranges reported for commercially relevant module classes, Equation (4) was applied to estimate the average point at which a module reaches 80% of its original generation capacity. In addition, Equation (3) was rearranged to determine the annual failure rate associated with this performance threshold, which was adopted here as the technical limit of the module’s functional lifetime. To parameterize energy production, a representative module from the Brazilian market was adopted as a reference case, namely the Canadian Solar HiKu6, a TOPCon module with an estimated service life of 25–35 years, weight of 27 kg, area of 2.2 m², fill form (FF) of 78%, and energy conversion efficiency (η) of 21.5%, which was among the best-selling solar panels in Brazil in 2023 [80–82]. This module was selected only as a reference case because of its market representativeness and the availability of technical data; it does not constitute a limitation of the proposed structure, which can be recalibrated for other manufacturers, technologies, degradation rates, and operating conditions.

Additionally, PV systems may generate energy in excess of instantaneous household demand (i.e., production > consumption), particularly during periods of high solar irradiance [78]. In this context, this surplus can be injected into the grid and converted into energy credits, which can be used to offset future consumption. To capture this dynamic,

the proposed framework moves beyond a purely threshold-based representation of energy adequacy and instead incorporates a utility-based approach, in which surplus generation contributes to consumer welfare through a marginal benefit function. This formulation allows for the representation of temporal mismatches between production and consumption, while accounting for the economic value associated with energy compensation mechanisms over time [11].

To characterize energy production over time, this study used the Brazilian Solar Energy Atlas from the Brazilian Institute for Space Research (INPE) to identify the average annual number of daily solar irradiation hours per region [77]. The Northeast region was selected as a reference because it presents the highest solar irradiance index, with an average of 5.8 h/day [77,78]. In addition, the region is highly relevant for distributed PV generation in Brazil, exceeding 5 GW of installed distributed solar capacity according to ANEEL/ABSOLAR data reported in 2024, and it also represents a broad territorial and demographic case, with approximately 1.55 million km² and more than 54 million inhabitants according to the 2022 Brazilian Demographic Census [83,84]. These characteristics make the Northeast a suitable reference case for evaluating distributed PV generation under high solar-resource conditions, while the model remains adaptable to other regions by changing the irradiation and demand parameters.

To estimate the number of modules required, the Canadian Solar HiKu6 model was selected as a reference, as it was one of the most widely sold models in the Brazilian market. According to the Brazilian standard (NBR), NBR 16690:2019 [79], a correction factor between 10% and 25% may be applied to account for losses inherent to PV systems. In this study, the upper limit of 25% was deliberately adopted as a conservative assumption, as it provides a greater allowance for system losses, including conversion losses, shading, and operational efficiency effects, and therefore avoids underestimating the required system capacity. The initial system power was calculated using Equation (2) at time zero ($t = 0$), and the minimum number of modules was determined using Equation (5), resulting in 30 PV modules required to meet the projected consumption.

4.1. Estimation of Consumer Satisfaction

To measure the utility of the PV system and consumer satisfaction over time, Equations (9) and (10) were applied. These equations express, respectively, the accumulated system utility (u_1) and the perceived system satisfaction (u_2), both derived from the difference between energy production and consumption. These functions capture the intertemporal dynamics of diminishing marginal utility, in accordance with microeconomic consumer theory [48,49].

The accumulated utility (u_1) represents the aggregate welfare gain associated with the generation surplus, increasing until break-even ($P = C$). Beyond this point, utility tends to stabilize. The perceived system satisfaction (u_2) reflects the extent to which the system meets energy demand, gradually declining as module degradation occurs and production efficiency decreases. Consequently, as technical performance deteriorates, consumers perceive lower marginal utility, reinforcing the microeconomic rationale for replacement or reuse decisions based on the cost–benefit trade-off.

This modelling approach links technical efficiency loss to the decline in economic satisfaction, quantifying the point at which the additional benefit of reuse equals the perceived cost of maintenance or replacement. Thus, the indicator u_2 becomes a dynamic behavioral parameter, essential for defining the simulated scenarios of PV module reuse and functional lifetime extension [42].

4.2. Simulation Model of Scenarios

After estimating energy production and consumption and measuring the accumulated utility (u_1) and perceived satisfaction (u_2) of the system, the next step was to identify the reference points required to construct the analysis scenarios. Three main milestones were defined (A, E, and R) representing, respectively, the point of attention to technical performance, the break-even, and the end-of-functional-life point. These milestones guide the formulation and comparison of the simulated scenarios, as described below:

- I. Point of Attention (A): the moment when the system, although still functional and advantageous, begins to show signs of performance degradation and declining satisfaction. Economically, this point is conceived as an early-warning indicator to support decisions regarding maintenance, replacement, reuse, or continued operation before a critical threshold is reached. In scenarios where the break-even point occurs before the end-of-functional-life limit, this milestone can be defined as the difference between milestone E and other parameters, such as the system's payback period. However, when milestone E occurs only after milestone R, milestone A must be referenced to R in order to preserve its preventive decision-support function.
- II. Break-even (E): defined as the moment when monthly energy production equals consumption. Technically, this marks the end of the system's energy self-sufficiency, and the consumer starts experiencing dissatisfaction with the PV system.
- III. Point of End-of-Functional-Life (R): the moment when the system reaches 80% of its original generation capacity, a technical limit used as a reference for extending the life of PV modules, according to NREL guidelines [42].

To map the range of decision-making strategies, we present both the technical and economic perspectives. Figure 3 illustrates these milestones through the joint evolution of technical and economic dimensions over time, structuring the four analytical scenarios described.

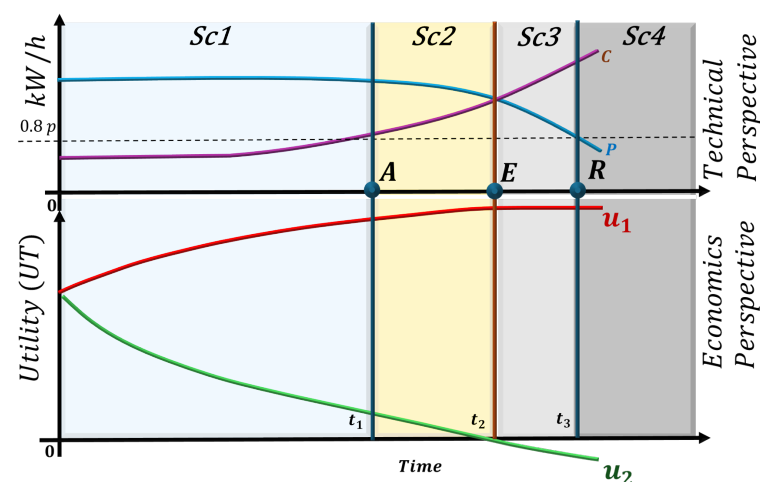


Figure 3. Mapping of the four analytical scenarios derived from the interaction between technical and economic dimensions over time. The technical dimension compares photovoltaic energy production (P) and household consumption (C), while the economic dimension represents accumulated utility (u_1) and marginal satisfaction (u_2). The temporal evolution of these variables defines three decision milestones: A, the attention point for anticipating maintenance, reuse, or replacement decisions; E, the break-even point at which production equals consumption ($P = C$); and R, the end-of-functional-life point associated with the 80% residual-generation threshold. The four scenarios represent, respectively, energy surplus with positive satisfaction, declining surplus approaching break-even, partial energy deficit with negative marginal satisfaction, and technical end-of-functional-life.

Scenario 1 (Sc_1) corresponds to a stage of energy oversupply, in which electricity production (P) remains higher than consumption (C), even though generation capacity gradually declines because of PV module degradation. At this stage, household demand may grow over time, but a surplus of electricity is still maintained, avoiding additional energy costs. From an economic perspective, the system remains advantageous because accumulated utility (u_1) continues to increase, while marginal utility (u_2) remains positive, indicating that the system still satisfies consumer demand. Under these conditions, there is no economic rationale for considering second-life.

Scenario 2 (Sc_2) represents the moment of attention, when the surplus between production and consumption begins to shrink significantly. Although production may still exceed consumption initially, this gap progressively narrows until the system reaches the break-even point (E), at which generation equals demand. Economically, accumulated utility (u_1) reaches its maximum, while marginal utility (u_2) approaches zero, signaling the onset of consumer dissatisfaction and indicating that the system no longer provides additional benefits relative to household needs.

Scenario 3 (Sc_3) describes a stage of partial energy deficit, in which consumption (C) exceeds production (P). At this point, surplus generation is no longer available, and the consumer becomes increasingly dependent on external electricity supply, with corresponding cost implications where compensation mechanisms are absent. In economic terms, accumulated utility (u_1) begins to decline, while marginal utility (u_2) becomes negative, continuing to drop over time. The system still operates but no longer fully meets demand.

Scenario 4 (Sc_4) corresponds to the end-of-functional-life stage. In this scenario, production remains below consumption and the economic behavior follows the declining pattern observed in Sc_3 . However, the defining feature of Sc_4 is that the system has reached the technical threshold of functional EoL, conventionally defined as 80% of its original generation capacity according to NREL guidelines. At this stage, the module is no longer considered suitable for reuse, and EoL management must therefore focus on alternative pathways such as recycling or final disposal.

4.3. Simulation of Consumers' Behavior

To analyze the interaction between technical–economic perspectives and consumption behaviors, we simulated a single production configuration using Class A as the reference case for the initial expansion phase of residential PV adoption, under three different consumption growth patterns: (i) Early: 1% per year, (ii) Moderate: 0.5% per year; and (iii) Stable: 0.05% per year. These interactions resulted in the construction of three behaviors (1, 2, and 3) and the identification of temporal milestones A, E, and R, illustrated in Figure 4.

The values of the simulation milestones are summarized in Table 2. Milestone R was estimated to occur at year 29.58, given that the estimated useful life of this equipment ranges from 25 to 35 years. Milestone E was determined by comparing Equations (1) and (2). For Behaviors 1 and 2, milestone A was defined as milestone E minus the system's payback period, because the production–consumption break-even occurs before the module reaches the 80% end-of-functional-life threshold. In Behavior 3, however, milestone E occurs only after milestone R. Under these conditions, premature disposal is not expected, and using E as the reference for A would reduce the practical value of A as an early-warning indicator. Therefore, for Behavior 3, milestone A was defined based on milestone R, with an assigned payback period of 2 years, preserving its role as a preventive decision-support signal before the module reaches the technical end-of-functional-life limit. Additional data are available in the Supplementary Material.

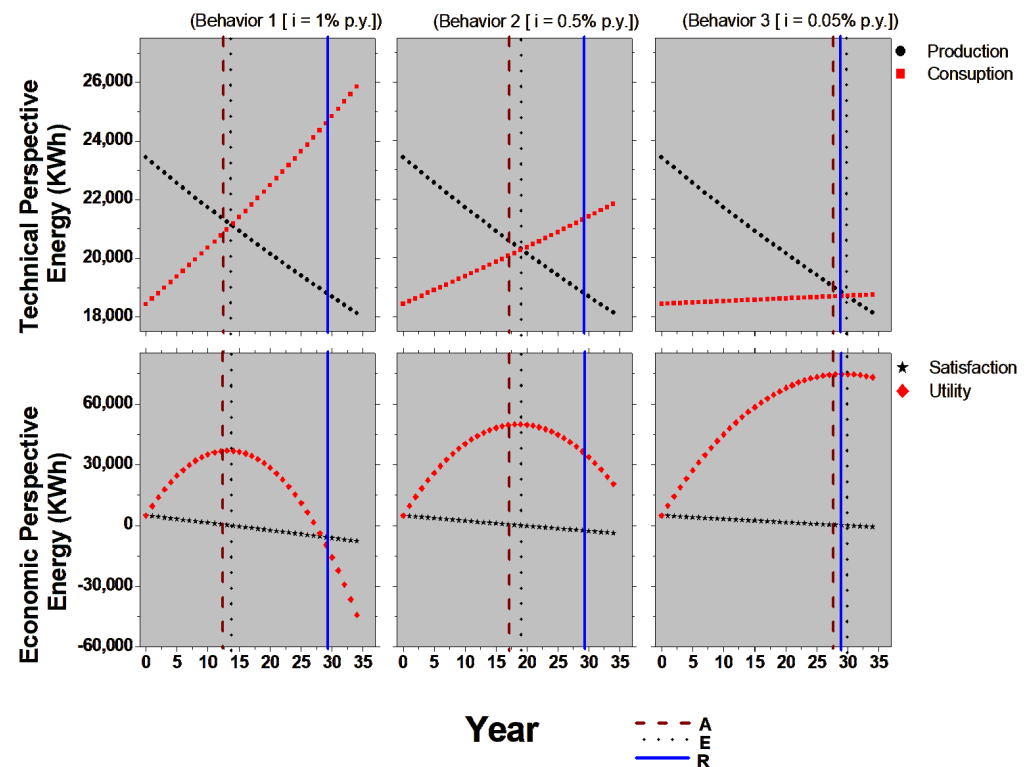


Figure 4. Computational simulation of three projected consumption-growth scenarios and their technical–economic effects. Behavior 1 represents early growth at 1% per year, Behavior 2 moderate growth at 0.5% per year, and Behavior 3 stable growth at 0.05% per year. These rates alter the production–consumption relationship over time and shift the milestones A, E, and R, indicating different moments for preventive attention, reuse or replacement assessment, and end-of-functional-life management.

Table 2. Simulation parameters and milestones for the three projected consumer behaviors. The table reports the annual consumption growth rate and the corresponding milestones A, E, and R. Behaviors 1 and 2 indicate potential premature-disposal pressure, whereas Behavior 3, due to its very low growth rate, reaches break-even only after R and therefore does not indicate premature disposal within the module’s functional lifetime.

Behavior	Growth Rate	A	E	R
1	1%	11.7	13.7	29.6
2	0.5%	17.2	19.2	29.6
3	0.05%	27.6	29.9	29.6

5. Discussion

The results suggest that the framework should be understood as a conceptual analytical structure focused on the relationship between production and consumption and on how this relationship may influence perceived satisfaction over time. Figure 3 illustrates this relationship along a shared time domain, where specific points (A, E, and R) can support objective decision-making based on predefined assumptions. These points may indicate appropriate moments for maintenance, system expansion, replacement, reuse assessment, or EoL planning. Furthermore, the framework highlights the dynamic relationship between production and consumption across different scenarios.

The three simulations (Figure 4) provide insights into how consumer profiles, represented by different consumption patterns, may influence the dynamics captured by the proposed framework. The interaction between economic and consumption perspectives

resulted in different values for the three temporal milestones (A—Alert, E—Break-even, and R—EoL). Figure 4 illustrates the three behavioral scenarios by combining technical and economic perspectives with three levels of consumption profiles. In Behavior 1, dissatisfaction occurs at approximately year 13.7 (see Table 2), suggesting that actions such as upgrading the PV system to restore increasing satisfaction could lead to the premature replacement of some modules, even though they remain approximately 16 years away from milestone R.

In the stable behavior scenario (Behavior 3 in Figure 4), unlike Behavior 1, the break-even point ($P = C$) and the transition to negative marginal utility ($u_2 < 0$) occur at around year 30, after the module has already reached the technical end-of-functional-life threshold. In this case, the consumer's electricity demand remains relatively stable, allowing the system's generation capacity to be utilized throughout most of its functional lifetime. Consequently, neither premature module replacement nor early system replacement is expected. This behavior represents a low-consumption-growth trajectory in which the economic trigger for replacement occurs after the technical degradation limit has been reached. Therefore, milestone A was defined relative to milestone R rather than milestone E, ensuring that the alert remains useful for anticipating maintenance, reuse assessment, or end-of-life (EoL) planning before the module reaches the 80% residual-capacity threshold.

Behaviors 1 and 3 represent the two extremes of consumer behavior considered in the simulations. Behavior 2, characterized by an intermediate electricity consumption growth rate of 0.5% per year, results in a warning milestone at approximately year 17, providing a two-year window for decision-making. If the consumer decides to upgrade the system at milestone E, approximately 12 years of remaining functional life would still be available for potential module reuse. Therefore, Behavior 2 illustrates a scenario in which a second useful life for PV modules is feasible, whereas this is not the case in Behavior 1, where the residual capacity has already fallen below the 80% threshold.

Figure 5 was developed from the scenarios described in Section 4.2. The framework relates photovoltaic production and household consumption over time to estimate the owner's level of satisfaction with the system. When production remains above consumption, the system delivers a surplus and the perceived satisfaction tends to remain positive, indicating no immediate need for intervention. As production declines and consumption increases, the reduction in the surplus signals an attention stage, in which the owner may begin to evaluate maintenance, replacement, or second-life alternatives for the original equipment. When consumption exceeds production, satisfaction decreases because the system no longer fully offsets household demand. Finally, in Scenario 4, the module reaches the technical end-of-functional-life threshold, with limited residual generation capacity; at this stage, second-life use is no longer considered feasible and end-of-life management should prioritize recycling or final disposal.

In emerging economies, such as Brazil, the rapid expansion of PV markets has occurred faster than the development of regulatory frameworks capable of supporting circular practices, including module reuse, refurbishment, and reverse logistics. The absence of Extended Producer Responsibility (EPR), dedicated reverse logistics schemes, technical standards, and certification mechanisms creates challenges for establishing reliable second-life markets and may contribute to the premature disposal of functional modules. In this context, complementary approaches that support decision-making throughout the PV system life cycle become particularly relevant. The proposed framework addresses this gap by integrating technical performance, economic conditions, and consumer behavior to identify appropriate moments for maintenance, replacement, reuse assessment, and end-of-life management. Although it does not replace regulatory instruments, the framework can provide valuable support for stakeholders by improving the visibility of reuse

opportunities and informing strategies for circular economy implementation in contexts where comprehensive policies are still emerging [85,86].

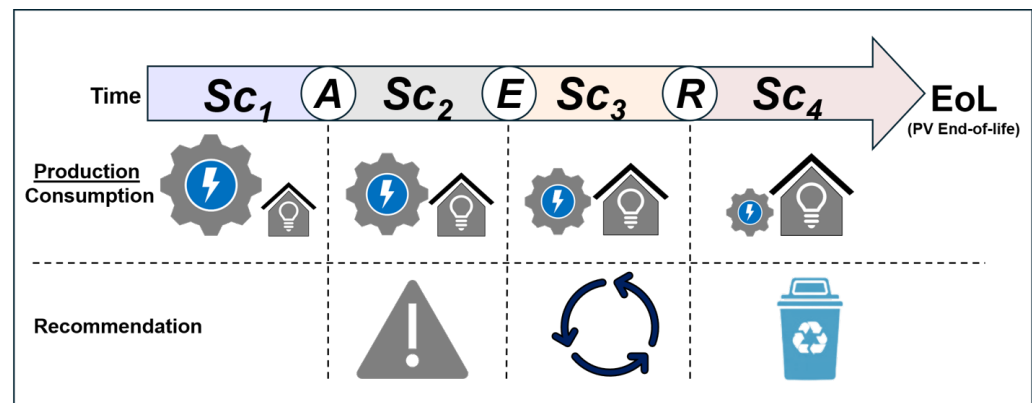


Figure 5. Consumer satisfaction across four analytical scenarios and three milestones (A, E, and R). The framework links photovoltaic production and household consumption over time to evaluate the evolution of perceived satisfaction, identify attention stages for maintenance, replacement, or second-life assessment, and distinguish cases in which residual generation capacity is insufficient for reuse and end-of-life management should prioritize recycling.

This regulatory gap limits the realization of circular economy benefits in rapidly expanding PV markets such as Brazil, where the absence of technical standards, certification procedures, and economic incentives may lead to the premature disposal of functional modules. Rather than replacing regulatory initiatives, the proposed framework offers a complementary decision-support perspective by integrating technical performance, economic viability, and consumer behavior into the assessment of PV system replacement and module reuse. Such an approach may help stakeholders identify opportunities for extending module service life while more comprehensive regulatory frameworks are still evolving. In addition, increasing consumer awareness of the relationship between electricity demand, system performance, and replacement decisions can encourage more sustainable choices, including the adoption of certified second-life modules where appropriate. Similar evidence has been reported in other sectors, where consumer decisions aligned with both economic preferences and environmental objectives have contributed to reducing greenhouse gas (GHG) emissions [87]. Consequently, combining consumer-oriented decision tools with the gradual development of EPR policies, reverse logistics systems, and certification standards may represent a practical pathway for advancing circular economy strategies in emerging countries [11,86,88].

Even as an intermediate approach, the adoption of second-life PV modules may provide environmental and resource-efficiency benefits by extending module service life and reducing the premature generation of photovoltaic waste. By supporting the identification of suitable opportunities for reuse and delaying end-of-life decisions, the proposed framework may be aligned with the objectives of United Nations Sustainable Development Goal 12.5, which seeks to substantially reduce waste generation through prevention, reduction, recycling, and reuse. In this context, second-life strategies may represent a potential pathway for promoting more sustainable consumption and production patterns in emerging PV markets.

Despite these potential contributions to more sustainable consumption and production patterns, the transition toward a circular photovoltaic sector still depends on overcoming important practical and operational barriers. The large-scale implementation of second-life strategies requires careful consideration of challenges associated with dismantling, performance testing, re-certification, and reverse logistics, which may involve transaction

costs and operational complexity. Addressing these barriers, together with the development of appropriate regulatory and market mechanisms, will be a condition for enabling the broader adoption of circular practices and realizing the potential benefits of PV module reuse [12,23,27].

6. Conclusions

This study proposed a technical–economic framework to support decision-making in the LCM of residential PV modules. The main contribution of the approach lies in integrating module degradation, household electricity-consumption dynamics, and decision-support criteria into a single parameterizable structure. Rather than introducing a new degradation model or proposing an alternative waste-management pathway, the framework provides a practical approach for identifying when changes in technical performance and household demand modify the energy balance and influence replacement-related decisions in residential PV systems.

The simulation results demonstrate that consumption-growth scenarios affect the timing of the proposed decision milestones. Under higher consumption-growth conditions, the attention and break-even milestones occur earlier, increasing the likelihood of premature replacement while modules may still retain substantial residual capacity. Conversely, under stable consumption conditions, the system remains functional for a longer period, and the economic trigger for replacement occurs closer to the technical end-of-functional-life threshold. These findings indicate that premature PV waste is not determined solely by module degradation but also by the interaction between declining generation capacity and evolving household electricity demand.

The proposed milestones provide a structured basis for anticipating decisions regarding continued operation, maintenance, system expansion, replacement, reuse assessment, and EoL planning. In practical terms, the framework can assist residential system owners, service providers, policymakers, and other stakeholders in evaluating whether modules should remain in operation, be considered for second-life applications, or proceed to recycling and other EoL pathways. Its parameterizable structure also enables adaptation to different regions, income contexts, module technologies, irradiation conditions, and decision thresholds, supporting broader life-cycle planning for distributed PV systems.

Beyond individual replacement decisions, the framework contributes to discussions on circular economy strategies for the PV sector by linking technical performance indicators with consumer-oriented decision points. By identifying situations in which modules may no longer meet the economic expectations of their original users while remaining technically suitable for continued use, the approach may support the development of second-life markets, reverse-logistics strategies, certification mechanisms, and complementary policy instruments aimed at reducing premature PV waste. In this sense, the framework provides a methodological basis for connecting residential decision-making with circular economy principles and more sustainable LCM of PV modules, particularly in markets where regulatory structures are still evolving.

This study has limitations that should guide future research. The degradation process was represented using a constant rate due to the absence of detailed degradation trajectories under the reference operating conditions, and the framework has not yet been empirically validated using monitored field data or observed consumer decisions. Future studies should incorporate non-linear, technology-specific, and climate-dependent degradation profiles, as well as additional economic, regulatory, and behavioral variables, including electricity tariffs, maintenance costs, incentives, financing conditions, risk perception, and willingness to adopt second-life modules. A direct international comparison was not conducted because a consistent application of the framework would require the collection

and harmonization of region-specific technical, climatic, economic, regulatory, market, and household-demand parameters. Future research should therefore recalibrate and apply the framework to European and Southeast Asian contexts, enabling cross-regional comparisons of premature PV replacement and identifying the factors that differentiate the Brazilian emerging market from more mature or climatically distinct PV markets. Empirical validation using operational PV data and real replacement, reuse, and disposal decisions will be essential to calibrate the proposed milestones and strengthen the framework's applicability for life-cycle planning, policy development, and stakeholder engagement in the PV sector.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/cleantechnol8050141/s1>, Sheet S1: Time-series data for production and consumption scenarios; Sheet S2: Temporal milestones (A, E, and R); Sheet S3: Dataset used to generate Figure 4; Table S1: MetaDate: Data dictionary describing all variables used in the study.

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Data Availability Statement: The data presented in this study are openly available in <https://github.com/GroupNPAI/Second-Life> (accessed on 19 July 2026).

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Conflicts of Interest: The authors declare that we have no financial, commercial, or personal conflicts of interest that could influence the results or interpretations presented in this manuscript.

Abbreviations

The following abbreviations are used in this manuscript:

ANEEL	Brazilian Electric Energy Agency
EoL	End-of-Life
EPE	Energy Research Office
EPR	Extended Producer Responsibility
EU	European Union
EVA	Ethylene-Vinyl Acetate
FF	Fill Factor
GHG	Greenhouse Gases
IEA	International Energy Agency
INPE	Brazilian Institute for Space Research
IoT	Internet of Things
LCM	Life-cycle management
LCOE	Levelized Cost of Energy
MTTF	Mean Time to Failure
NBR	Brazilian Standard

NREL	National Renewable Energy Laboratory
PV	Photovoltaic
SDG	Sustainable Development Goals
WEEE	Waste of Electrical and Electronic Equipment

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